



1/22

经典电动力学

清华大学物理系

王 青



Back

Close



2/22

习题课一



Back

Close



3/22

$$\sum_{k=1}^3 \epsilon_{ijk} \epsilon_{lmk} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl} \quad (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \quad \text{习题}$$

$$= (\vec{e}_i \times \vec{e}_j) \cdot (\vec{e}_l \times \vec{e}_m) = (\vec{e}_i \cdot \vec{e}_l)(\vec{e}_j \cdot \vec{e}_m) - (\vec{e}_i \cdot \vec{e}_m)(\vec{e}_j \cdot \vec{e}_l)$$

$$\int_{\text{包括 } \mathbf{x}_0} d\mathbf{x} \, g(\mathbf{x}) \delta(\mathbf{x} - \mathbf{x}_0) = g(\mathbf{x}_0)$$

$$\delta(a\mathbf{x}) = \frac{1}{a} \delta(\mathbf{x}) \quad a > 0 \quad a < 0? \quad \delta(\mathbf{f}(\mathbf{x})) = \sum_i \frac{1}{|\mathbf{f}'(\mathbf{x}_i)|} \delta(\mathbf{x} - \mathbf{x}_i)$$

$$\mathbf{x} \delta(\mathbf{x}) = 0$$

$$\int_{\text{包括 } 0} d\mathbf{x} \, g(\mathbf{x}) \delta(a\mathbf{x}) \stackrel{y=a\mathbf{x}}{=} \frac{1}{a} \int_{\text{包括 } 0} d\mathbf{y} \, g\left(\frac{\mathbf{y}}{a}\right) \delta(\mathbf{y}) = \frac{1}{a} g(0) = \int_{\text{包括 } 0} d\mathbf{x} \, g(\mathbf{x}) \frac{1}{a} \delta(\mathbf{x})$$

$$a < 0 \quad \delta(a\mathbf{x}) = \delta(-a\mathbf{x}) = -\frac{1}{a} \delta(\mathbf{x}) \quad \text{一般地: } \delta(a\mathbf{x}) = \frac{1}{|a|} \delta(\mathbf{x})$$

$$\int_{-\infty}^{\infty} d\mathbf{x} \, g(\mathbf{x}) \delta(\mathbf{f}(\mathbf{x})) = \sum_i \int_{x_i \text{ 的邻域}} d\mathbf{x} \, g(\mathbf{x}_i) \delta(\mathbf{f}(\mathbf{x})) = \sum_i \int_{x_i \text{ 的邻域}} d\mathbf{f} \frac{g(\mathbf{x}_i)}{|\mathbf{f}'(\mathbf{x})|} \delta(\mathbf{f})$$

$$= \sum_i \frac{g(\mathbf{x}_i)}{|\mathbf{f}'(\mathbf{x}_i)|} = \int_{-\infty}^{\infty} d\mathbf{x} \, g(\mathbf{x}) \sum_i \frac{1}{|\mathbf{f}'(\mathbf{x}_i)|} \delta(\mathbf{x} - \mathbf{x}_i)$$

$$\int_{\text{包括 } 0} d\mathbf{x} \, g(\mathbf{x}) \mathbf{x} \delta(\mathbf{x}) = g(\mathbf{x}) \mathbf{x} \Big|_{\mathbf{x}=0} = 0$$



Back

Close



数学准备矢量分析补充题(c)思考选作题:

1. 一阶张量除零矢量外, 都不是各向同性的
2. 二阶张量中的各向同性的张量必为 $\lambda \delta_{ij}$
3. 三阶张量中的各向同性的张量必为 $\lambda \epsilon_{ijk}$
4. 四阶张量中的各向同性的张量必为 $\lambda \delta_{ij} \delta_{kl} + \alpha \delta_{ik} \delta_{jl} + \beta \delta_{il} \delta_{jk}$

考虑一个非零各向同性矢量 $\vec{v}$ 将 x 轴的方向取在 $\vec{v}$ 的方向 $\vec{v} = (v, 0, 0)$

将坐标系转到 $-x$ 方向, 在新的坐标架下: $\vec{v} = (-v, 0, 0)$

$$v = -v \Rightarrow v = 0$$



Back

Close



二阶张量中的各向同性的张量必为 $\lambda\delta_{ij}$.

考虑一个各向同性二阶张量 $\mathbf{A}_{ij}$ 将 x, y, z 轴的取向任意性导致

$$\mathbf{A}_{11} = \mathbf{A}_{1'1'} = \mathbf{A}_{22} = \mathbf{A}_{33} \quad \mathbf{A}_{12} = \mathbf{A}_{23} = \mathbf{A}_{31} \quad \mathbf{A}_{21} = \mathbf{A}_{32} = \mathbf{A}_{13}$$

$$\mathbf{A}'_{ij} = \sum_{i'j'} \beta_{ii'} \beta_{jj'} \mathbf{A}_{i'j'}$$

坐标架绕 z 轴转 180° : $\mathbf{x}'_1 = -\mathbf{x}_1 \quad \mathbf{x}'_2 = -\mathbf{x}_2 \quad \mathbf{x}'_3 = \mathbf{x}_3$

$$\mathbf{A}_{23} = \mathbf{A}'_{23} = \sum_{mn} \beta_{2m} \beta_{3n} \mathbf{A}_{mn} = -\mathbf{A}_{23}$$

$$\mathbf{A}_{12} = \mathbf{A}_{23} = \mathbf{A}_{31} = \mathbf{A}_{21} = \mathbf{A}_{32} = \mathbf{A}_{13} = 0 \quad \mathbf{A}_{ij} = \lambda\delta_{ij}$$

充分性:
$$\mathbf{A}'_{ij} = \sum_{i'j'} \beta_{ii'} \beta_{jj'} \mathbf{A}_{i'j'} = \sum_{i'j'} \beta_{ii'} \beta_{jj'} \lambda\delta_{i'j'} = \lambda \sum_{i'} \underbrace{\beta_{ii'} \beta_{ji'}}_{\delta_{ij}} = \lambda\delta_{ij}$$



Back

Close

三阶张量中的各向同性的张量必为 $\lambda \epsilon_{ijk}$

习题



6/22

$$\mathbf{A}_{ijk} = \mathbf{A}'_{ijk} = \sum_{i'j'} \beta_{ii'} \beta_{jj'} \beta_{kk'} \mathbf{A}_{i'j'k'} \quad \text{考虑 } (\beta_{ij}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A_{111} = A'_{111} = \beta_{11}^3 A_{111} = -A_{111} \Rightarrow A_{111} = A_{222} = A_{333} = 0$$

i, j, k 中有两个3, 一个1或2: $\mathbf{A}_{ijk} = -\mathbf{A}_{ijk} \Rightarrow$ 若有两个角标相同, 则为零!

将 x, y, z 轴的取向任意性导致: $\mathbf{A}_{123} = \mathbf{A}_{231} = \mathbf{A}_{312} \quad \mathbf{A}_{213} = \mathbf{A}_{231} = \mathbf{A}_{132}$

$$\text{绕 } z \text{ 轴转 } 90^\circ \quad (\beta_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{A}_{123} = \mathbf{A}'_{123} = \sum_{mnl} \beta_{1m} \beta_{2n} \beta_{3l} \mathbf{A}_{mnl} = \beta_{12} \beta_{21} \beta_{33} \mathbf{A}_{213} = -\mathbf{A}_{213}$$

充分性: $\mathbf{A}'_{ijk} = \sum_{i'j'k'} \beta_{ii'} \beta_{jj'} \beta_{kk'} \mathbf{A}_{i'j'k'} = \sum_{i'j'k'} \beta_{ii'} \beta_{jj'} \beta_{kk'} \lambda \epsilon_{i'j'k'} = \lambda (\det \beta) \epsilon_{ijk}$



Back

Close



四阶张量中的各向同性的张量必为 $\lambda \delta_{ij} \delta_{kl} + \alpha \delta_{ik} \delta_{jl} + \beta \delta_{il} \delta_{jk}$

$$A_{ijkl} = A'_{ijkl} = \sum_{mnpq} \beta_{im} \beta_{jn} \beta_{kp} \beta_{lq} A_{mnpq}$$

$$\text{绕} z \text{轴转} 180^\circ (\beta_{ij}) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

i, j, k, l 中有一个 3; m, n, p, q 中有一个 3: $A_{ijkl} = -A_{ijkl} = 0$ ■

$\Rightarrow i, j, k, l$ 中有单数个 1 或 2 或 3, 则相应的分量为零! ■

剩下: $i = j = k = l$ $i = j \neq k = l$ $i = k \neq j = l$ $i = l \neq k = j$ ■

$$A_{1111} = A_{2222} = A_{3333} \quad \text{绕} z \text{轴转} 90^\circ (\beta_{ij}) = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} A_{1122} &= A'_{1122} = A_{2211} & A_{1212} &= A_{2121} & A_{1221} &= A_{2112} \\ &= A_{2233} = A_{3311} & &= A_{2323} = A_{3131} & &= A_{2332} = A_{3113} \end{aligned}$$



Back

Close



四阶张量中的各向同性的张量必为 $\lambda\delta_{ij}\delta_{kl} + \alpha\delta_{ik}\delta_{jl} + \beta\delta_{il}\delta_{jk}$

$$\mathbf{A}_{ijkl} = \begin{cases} \gamma & \mathbf{i} = \mathbf{j} = \mathbf{k} = \mathbf{l} \\ \lambda & \mathbf{i} = \mathbf{j} \neq \mathbf{k} = \mathbf{l} \\ \alpha & \mathbf{i} = \mathbf{l} \neq \mathbf{j} = \mathbf{k} \\ \beta & \mathbf{i} = \mathbf{k} \neq \mathbf{l} = \mathbf{j} \\ 0 & \text{其它} \end{cases} \quad (\beta_{ij}) \stackrel{\text{绕}z\text{轴转}45^\circ}{=} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \gamma &= \mathbf{A}_{1111} = \mathbf{A}'_{1111} = \sum_{mnpq} \beta_{1m}\beta_{1n}\beta_{1p}\beta_{1q}\mathbf{A}_{mnpq} \\ &= \frac{1}{4}[\mathbf{A}_{1111} + \mathbf{A}_{2222} + \mathbf{A}_{1122} + \mathbf{A}_{2211} + \mathbf{A}_{1212} + \mathbf{A}_{2121} + \mathbf{A}_{1221} + \mathbf{A}_{2112}] \\ &= \frac{1}{4}[2\gamma + 2\lambda + 2\alpha + 2\beta] \Rightarrow \gamma = \lambda + \alpha + \beta \end{aligned}$$

$$\mathbf{A}_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \alpha\delta_{ik}\delta_{jl} + \beta\delta_{il}\delta_{jk}$$

充分性:
$$\begin{aligned} \mathbf{A}'_{ijkl} &= \sum_{i'j'k'} \beta_{ii'}\beta_{jj'}\beta_{kk'}\beta_{ll'} [\lambda\delta_{i'j'}\delta_{k'l'} + \alpha\delta_{i'k'}\delta_{j'l'} + \beta\delta_{i'l'}\delta_{j'k'}] \\ &= \lambda\delta_{ij}\delta_{kl} + \alpha\delta_{ik}\delta_{jl} + \beta\delta_{il}\delta_{jk} \end{aligned}$$



Back

Close



大纲第5章4(b)，证明n阶张量收缩得到n-2阶张量。

不少同学不知如何下手，不去证明收缩之后的量满足张量变换律，而只能含糊其辞地说”收缩之后指标变为n-2个”之类的话。■

$$\mathbf{T}'_{\mu_1 \cdots \mu_n} = \sum_{\nu_1, \cdots, \nu_n=1}^4 \mathbf{a}_{\mu_1 \nu_1} \cdots \mathbf{a}_{\mu_n \nu_n} \mathbf{T}_{\nu_1 \cdots \nu_n}$$

$$\sum_{\mu_i, \mu_j=1}^4 \delta_{\mu_i \mu_j} \mathbf{T}'_{\mu_1 \cdots \mu_i \cdots \mu_j \cdots \mu_n} = \sum_{\mu_i, \mu_j=1}^4 \delta_{\mu_i \mu_j} \sum_{\nu_1, \cdots, \nu_n=1}^4 \mathbf{a}_{\mu_1 \nu_1} \cdots \mathbf{a}_{\mu_n \nu_n} \mathbf{T}_{\nu_1 \cdots \nu_n}$$

$$\sum_{\mu_i, \mu_j=1}^4 \delta_{\mu_i \mu_j} \mathbf{a}_{\mu_i \nu_i} \mathbf{a}_{\mu_j \nu_j} = \delta_{\nu_i \nu_j}$$



Back

Close



习题

第一章第三节补充题(c): 假定实验上发现库伦定律对平方反比有一微小的指数偏离 δ , $\vec{f}_{21} = k_1 \frac{q_1 q_2}{R^3} \vec{R} e^{-\delta \frac{R}{R_0}}$. 求证, 关系 $\vec{E} = -\nabla \phi$ 仍成立, 并请对给定的电荷密度分布, 给出 ϕ 的计算公式。■

$$-\nabla \phi \stackrel{?}{=} k_1 \frac{q}{R^3} \vec{R} e^{-\delta \frac{R}{R_0}} \stackrel{\phi=\phi[\mathbf{R}]}{\implies} \nabla \phi[\mathbf{R}] = \phi'[\mathbf{R}] \frac{\vec{R}}{R} \quad \blacksquare$$

$$\phi'[\mathbf{R}] = -k_1 \frac{q}{R^2} e^{-\delta \frac{R}{R_0}} \quad \blacksquare$$

指数积分: $\text{Ei}(x) = \int_{-\infty}^x du \frac{e^u}{u} \quad \blacksquare = \int_{-\infty}^x \frac{de^u}{e^u} = \frac{e^x}{x} + \int_{-\infty}^x du \frac{e^u}{u^2} \quad \blacksquare$

$$\frac{d}{dx} \left[\text{Ei}(x) - \frac{e^x}{x} \right] = \frac{e^x}{x^2} \quad \Rightarrow \quad \frac{d}{dx} \left[\text{Ei}(ax) - \frac{e^{ax}}{ax} \right] = \frac{e^{ax}}{ax^2} \quad \blacksquare$$

$$a = -\frac{\delta}{R_0} \quad \Rightarrow \quad \frac{d}{dR} \left[\text{Ei}\left(-\delta \frac{R}{R_0}\right) + \frac{R_0 e^{-\delta \frac{R}{R_0}}}{\delta R} \right] = -\frac{R_0 e^{-\delta \frac{R}{R_0}}}{\delta R^2} \quad \blacksquare$$

$$\phi[\mathbf{R}] = -k_1 q \left[\frac{\delta}{R_0} \text{Ei}\left(-\delta \frac{R}{R_0}\right) + \frac{e^{-\delta \frac{R}{R_0}}}{R} \right]$$



Back

Close

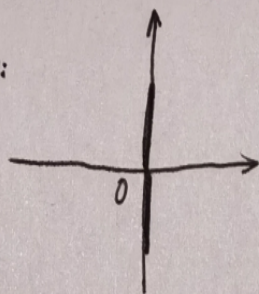


2.14 画出函数 $\frac{d\delta(x)}{dx}$ 的图,说明 $\rho = -(\vec{p} \cdot \nabla)\delta(\vec{r})$ 是一个位于原点的偶极子的电荷密度. ■

有9个同学误用了 $\nabla' = -\nabla$ 给出了一些奇怪的证明:

6. 书 2.14 画 $\frac{d\delta(x)}{dx}$ 的图 说明 $\rho = -(\vec{p} \cdot \nabla)\delta(\vec{r})$ 是一个位于原点的偶极子的电荷密度

解:



$$\vec{p} = q\vec{l} \quad +q, -q \text{ 位于 } +\frac{l}{2}, -\frac{l}{2} \text{ 处}$$

$$(\Delta x' e_x + \Delta y' e_y + \Delta z' e_z)$$

$$\text{当 } l \rightarrow 0 \quad \text{电荷密度 } \rho = q\delta(x - \frac{l}{2}) - q\delta(x + \frac{l}{2})$$

$$= q\Delta\delta(x)$$

$$= q \left[\frac{\partial\delta(\vec{x})}{\partial x'} \cdot \Delta x' + \frac{\partial\delta(\vec{x})}{\partial y'} \cdot \Delta y' + \frac{\partial\delta(\vec{x})}{\partial z'} \cdot \Delta z' \right]$$

$$= q\vec{l} \cdot \nabla'\delta(\vec{x}) = -\vec{p} \cdot \nabla\delta(\vec{x}) \quad (\nabla' = -\nabla) \text{ 得证}$$

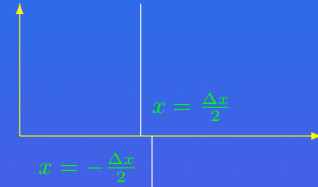


Back

Close



$$\frac{d\delta(\mathbf{x})}{d\mathbf{x}} = \lim_{\Delta\mathbf{x} \rightarrow 0} \frac{\delta(\mathbf{x} + \frac{\Delta\mathbf{x}}{2}) - \delta(\mathbf{x} - \frac{\Delta\mathbf{x}}{2})}{\Delta\mathbf{x}}$$



助教：相当一部分人是证明要证的电荷密度的偶极矩和偶极子的偶极矩一样！

$$\int dV \rho = - \int dV \mathbf{p}_i \partial_i \delta(\vec{\mathbf{r}}) = 0$$

$$\int dV x_i \rho = - \int dV x_i p_j \partial_j \delta(\vec{\mathbf{r}}) = - \int dV \{ \partial_j [x_i p_j \delta(\vec{\mathbf{r}})] - p_i \delta(\vec{\mathbf{r}}) \} = p_i$$

$$\begin{aligned} \int dV x_i x_j \rho &= - \int dV x_i x_j p_k \partial_k \delta(\vec{\mathbf{r}}) \\ &= - \int dV \{ \partial_k [x_i x_j p_k \delta(\vec{\mathbf{r}})] - (p_i x_j + p_j x_i) \delta(\vec{\mathbf{r}}) \} = 0 \end{aligned}$$

$$\int dV x_{i_1} x_{i_2} \cdots x_{i_n} \rho = 0 \quad n \geq 2$$



Back

Close



1.5 证明电偶极矩的变化率 $\frac{d\vec{p}}{dt} = \int dV \vec{j}(\vec{r}, t)$

$$\begin{aligned}
 \frac{d\vec{p}}{dt} &= \int dV \vec{r} \frac{\partial}{\partial t} \rho(\vec{r}, t) = - \int dV \vec{r} \nabla \cdot \vec{j}(\vec{r}, t) \\
 &= -\vec{e}_i \int dV x_i \partial_k j_k(\vec{r}, t) = \vec{e}_i \int dV \{ -\partial_k [x_i j_k(\vec{r}, t)] + j_i(\vec{r}, t) \} \\
 &= \int dV \vec{j} \\
 &= \int dV \{ -\nabla \cdot (\vec{j} \vec{r}) + \vec{j} \cdot \nabla \vec{r} \} \\
 &\stackrel{?}{=} \int dV \{ -\nabla (\vec{j} \cdot \vec{r}) + (\nabla \vec{r}) \cdot \vec{j} \} \\
 &\stackrel{?}{=} \int dV \left[\vec{r} \frac{\partial}{\partial t} \rho(\vec{r}, t) + \vec{\nabla} \rho(\vec{r}, t) \right] = \int dV [-\vec{r} \nabla \cdot \vec{j} + \vec{j}] \\
 &= \int dV [-\nabla \cdot (\vec{j} \vec{r}) + 2\vec{j}]
 \end{aligned}$$



Back

Close

习题

1.10 证明两个电流线圈的相互作用大小相等，方向相反。

恒定电流元 $\mathbf{J}_2 d\vec{l}_2$ 对 $\mathbf{J}_1 d\vec{l}_1$ 的作用力为



$$\vec{\mathbf{F}}_{21} = k_2 \oint_2 \oint_1 \mathbf{J}_1 d\vec{l}_1 \times \left(\frac{\mathbf{J}_2 d\vec{l}_2 \times \vec{\mathbf{R}}}{R^3} \right)$$

$$\vec{\mathbf{R}} = \vec{\mathbf{r}}_1 - \vec{\mathbf{r}}_2$$

$$= \oint_2 \oint_1 \mathbf{J}_1 d\vec{l}_1 \times \mathbf{J}_2 d\vec{l}_2 \times \frac{\vec{\mathbf{R}}}{R^3}$$

$$= - \oint_2 \oint_1 \mathbf{J}_2 d\vec{l}_2 \times \mathbf{J}_1 d\vec{l}_1 \times \frac{\vec{\mathbf{R}}}{R^3}$$

$$= -k_2 \oint_2 \oint_1 \mathbf{J}_2 d\vec{l}_2 \times \left(\frac{\mathbf{J}_1 d\vec{l}_1 \times \vec{\mathbf{R}}}{R^3} \right) = -\vec{\mathbf{F}}_{12}$$

$$\vec{\mathbf{A}} \times (\vec{\mathbf{B}} \times \vec{\mathbf{C}}) \neq -\vec{\mathbf{B}} \times (\vec{\mathbf{A}} \times \vec{\mathbf{C}})$$



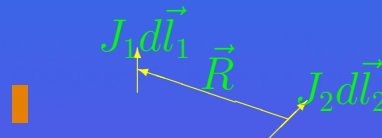
Back

Close

习题

1.10 证明两个电流线圈的相互作用大小相等，方向相反。

恒定电流元 $J_2 d\vec{l}_2$ 对 $J_1 d\vec{l}_1$ 的作用力为



$$\begin{aligned}\vec{F}_{21} &= k_2 \oint_2 \oint_1 J_1 d\vec{l}_1 \times \left(\frac{J_2 d\vec{l}_2 \times \vec{R}}{R^3} \right) & \vec{R} = \vec{r}_1 - \vec{r}_2 \\ &= \oint_2 \oint_1 \frac{k_2 J_1 J_2}{R^3} [(\vec{R} \cdot d\vec{l}_1) d\vec{l}_2 - (d\vec{l}_1 \cdot d\vec{l}_2) \vec{R}]\end{aligned}$$

电流元 $J_1 d\vec{l}_1$ 对 $J_2 d\vec{l}_2$ 的作用力为: $1 \leftrightarrow 2; \vec{R} \leftrightarrow -\vec{R}$

$$\begin{aligned}\oint_2 \oint_1 \frac{k_2 J_1 J_2}{R^3} (\vec{R} \cdot d\vec{l}_1) d\vec{l}_2 &= \oint_{2'} \oint_{1'} \frac{k_2 J_{1'} J_{2'}}{|\vec{r}_{1'} - \vec{r}_{2'}|^3} (\vec{r}_{1'} - \vec{r}_{2'}) \cdot d\vec{l}_{1'} d\vec{l}_{2'} \\ &\stackrel{1'=2, 2'=1}{=} \oint_1 \oint_2 \frac{k_2 J_1 J_2}{R^3} (-\vec{R} \cdot d\vec{l}_2) d\vec{l}_1\end{aligned}$$



Back

Close



1.10 证明两个电流线圈的相互作用大小相等，方向相反。

恒定电流元 $J_2 d\vec{l}_2$ 对 $J_1 d\vec{l}_1$ 的作用力为



$$\begin{aligned}\vec{F}_{21} &= k_2 \oint_2 \oint_1 J_1 d\vec{l}_1 \times \left(\frac{J_2 d\vec{l}_2 \times \vec{R}}{R^3} \right) \quad \vec{R} = \vec{r}_1 - \vec{r}_2 \\ &= \oint_2 \oint_1 \frac{k_2 J_1 J_2}{R^3} [(\vec{R} \cdot d\vec{l}_1) d\vec{l}_2 - (d\vec{l}_1 \cdot d\vec{l}_2) \vec{R}] \text{ber } \vec{R}\end{aligned}$$

电流元 $J_1 d\vec{l}_1$ 对 $J_2 d\vec{l}_2$ 的作用力为: $1 \leftrightarrow 2; \vec{R} \leftrightarrow -\vec{R}$

$$\begin{aligned}\oint_2 \oint_1 \frac{k_2 J_1 J_2}{R^3} (\vec{R} \cdot d\vec{l}_1) d\vec{l}_2 &= -k_2 \oint_1 J_2 d\vec{l}_2 \oint_1 J_1 d\vec{l}_1 \cdot \nabla_1 \frac{1}{R} \\ &= -k_2 \int dV_2 \vec{j}_2 \int dV_1 \left(\nabla_1 \frac{1}{R} \right) \cdot \vec{j}_1 \quad \text{闭合回路} \\ &= -k_2 \int dV_2 \vec{j}_2 \int dV_1 \left\{ [\nabla_1 \cdot \frac{\vec{j}_1}{R}] - \frac{1}{R} \nabla_1 \cdot \vec{j}_1 \right\} = 0\end{aligned}$$

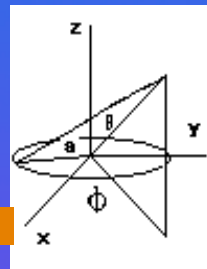
如果 $J_1 = J_2$ 则得到 闭合电流圈 无自作用 !



Back

Close

第一章第三节补充作业(b): 求证磁偶极子(电流圈面积 $S \rightarrow 0$, 电流 $J \rightarrow \infty$, JS 固定)产生的矢量势为,



$$\vec{A} = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{r}}{r^3} + \text{常数} \quad \vec{S} = \frac{1}{2} \oint \vec{a} \times d\vec{l} \quad \vec{m} = J\vec{S}$$

$$\vec{A} = k_2 \oint d\vec{l} \frac{J}{|\vec{r} - \vec{a}|} = k_2 \int d\vec{S}_a \times \nabla_a \frac{J}{|\vec{r} - \vec{a}|} = k_2 \int d\vec{S}_a \times \frac{J\vec{r}}{r^3}$$

$$\vec{A} = k_2 \oint d\vec{l} \frac{J}{|\vec{r} - \vec{a}|} = k_2 \oint d\vec{l} \frac{J}{\sqrt{r^2 - 2\vec{r} \cdot \vec{a}}} = k_2 \oint d\vec{l} \frac{J}{r \sqrt{1 - 2\vec{r} \cdot \vec{a}/r^2}}$$

$$= k_2 \oint d\vec{l} \frac{J}{r} \left[1 + \frac{\vec{r} \cdot \vec{a}}{r^2} \right] = \frac{k_2 J}{r^3} \oint d\vec{l} \vec{r} \cdot \vec{a} = \frac{k_2 J}{2r^3} \oint [d\vec{l} \vec{r} \cdot \vec{a} - \vec{a} d\vec{l} \cdot \vec{r}]$$

$$= \frac{k_2 J}{2r^3} \oint (\vec{a} \times d\vec{l}) \times \vec{r} = \frac{k_2 J}{r^3} \vec{S} \times \vec{r} = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \vec{r}}{r^3}$$

$$\oint J [d\vec{l} \vec{r} \cdot \vec{a} + \vec{a} d\vec{l} \cdot \vec{r}] = \int d\tau_a [\vec{j}(\vec{a}) \vec{r} \cdot \vec{a} + \vec{a} \vec{j}(\vec{a}) \cdot \vec{r}] = \int d\tau_a \nabla_a \cdot [\vec{j}(\vec{a}) \vec{a} \vec{r} \cdot \vec{a}]$$

$$= \int d\tau_a [(\nabla \cdot \vec{j}) \vec{a} \vec{r} \cdot \vec{a} + \vec{j} \cdot (\nabla \vec{a}) \vec{r} \cdot \vec{a} + \vec{a} \vec{j} \cdot (\nabla \vec{a}) \cdot \vec{r}] = 0$$





第一章第14题：内外半径分别为a和b的无限长圆柱形电容器，单位长度电荷为 λ_f ，板间填充电导率为 σ 的非磁性物质。

(1)证明在介质中任何一点传导电流与位移电流严格抵消，因此内部无磁场。

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0 \Rightarrow \frac{\partial}{\partial t}(\nabla \cdot \vec{D}) + \nabla \cdot \vec{j} = 0 \Rightarrow \nabla \cdot \left[\frac{\partial \vec{D}}{\partial t} + \vec{j} \right] = 0$$

$$\Rightarrow \frac{\partial \vec{D}}{\partial t} + \vec{j} \equiv \Delta \vec{j} \stackrel{???}{=} 0 \quad \Leftarrow \quad \text{它实际只告诉我们 } \Delta \vec{j} \text{ 是稳恒电流!}$$

$$\Delta \vec{j} = \Delta j_r \vec{e}_r + \Delta j_\theta \vec{e}_\theta + \Delta j_z \vec{e}_z$$

系统均匀带电无限长及旋转对称： $\Delta j_z = 0$ $\Delta j_\theta = 0$ Δj_r 不依赖坐标 θ, z

做个半径r高为l的柱状高斯面： $0 = \oint d\vec{S} \cdot \Delta \vec{j} = 2\pi r l \Delta j_r$



Back

Close



1. 积分形式与微分形式的Maxwell方程组哪个更普遍，为什么？

2. 证明介电常数为无穷的电介质表面为等势面

- 积分形式是有限远的场之间的关联，微分形式是无穷靠近的场之间的关联
- 有限远显然包含无穷靠近，因而积分形式包含微分形式
- 虽然将无穷多的无穷靠近累积起来可以得到有限远，但微分形式存在要求较高
- 微分形式无法完全囊括积分形式，因而平凡拓扑下积分形式更普遍
- 但积分区域在某些情形下不能很好定义，这时反而是微分形式可以应用
- $\vec{D} = \epsilon \vec{E}$ 怎么证明 $\vec{D}$ 在介电常数为无穷时是有限的？
- 在介电常数无穷的介质上的方程 $\nabla^2 \phi = -\rho_f / \epsilon = 0$ 。但这不足以说明就是导体！
- 边界上： $\phi \Big|_{\text{介电常数无穷的介质边界}} = \phi' \Big|_{\text{其它介质边界}} ; \epsilon \frac{\partial \phi}{\partial n} \Big|_{\text{介电常数无穷的介质边界}} = \epsilon' \frac{\partial \phi'}{\partial n} \Big|_{\text{其它介质边界}}$
- 碰到与前面类似的问题：如何证明 $\frac{\partial \phi'}{\partial n} \Big|_{\text{其它介质边界}}$ 在介电常数为无穷时是有限的？



Back

Close



相对论力学题 (c) : 从 $K_\mu = \frac{dp_\mu}{d\tau}$ 出发, 证明 $K_4 = \frac{i}{c} \vec{K} \cdot \vec{v}$

很多同学用了电磁力的表达式: $K_\mu = e \sum_\nu F_{\mu\nu} U_\nu$

$$\begin{aligned}
 K_i &= e \sum_\nu F_{i\nu} U_\nu = e \sum_j F_{ij} U_j + e F_{i4} U_4 = e \sum_{jk} \epsilon_{ijk} U_j B_k - e \frac{i}{c} E_i U_4 \\
 &= e \sum_{jk} \epsilon_{ijk} \frac{v_j B_k}{\sqrt{1 - \frac{v^2}{c^2}}} - e \frac{i}{c} \frac{ic E_i}{\sqrt{1 - \frac{v^2}{c^2}}} \\
 K_4 &= e \sum_\nu F_{4\nu} U_\nu = e \sum_i F_{4i} U_i = e \sum_i \frac{i}{c} E_i \frac{v_i}{\sqrt{1 - \frac{v^2}{c^2}}} = \sum_i \frac{i}{c} K_i v_i = \frac{i}{c} \vec{K} \cdot \vec{v}
 \end{aligned}$$

而题意是要从定义出发, 不限电磁作用

$$\begin{aligned}
 K_i &= \frac{dp_i}{d\tau} = \frac{d \frac{m_0 v_i}{\sqrt{1 - \frac{v^2}{c^2}}}}{d\tau} = \frac{m_0 \frac{dv_i}{d\tau}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{1}{2} m_0 v_i \frac{\frac{d}{d\tau} (-\frac{v^2}{c^2})}{(1 - \frac{v^2}{c^2})^{\frac{3}{2}}} \\
 \vec{K} \cdot \vec{v} &= \frac{\frac{1}{2} m_0 \frac{dv^2}{d\tau}}{\sqrt{1 - \frac{v^2}{c^2}}} + \frac{1}{2} m_0 \frac{v^2}{c^2} \frac{\frac{dv^2}{d\tau}}{(1 - \frac{v^2}{c^2})^{\frac{3}{2}}} = \frac{\frac{1}{2} m_0 \frac{dv^2}{d\tau}}{\sqrt{1 - \frac{v^2}{c^2}}} \left[1 + \frac{\frac{v^2}{c^2}}{1 - \frac{v^2}{c^2}} \right] = \frac{\frac{1}{2} m_0 \frac{dv^2}{d\tau}}{(1 - \frac{v^2}{c^2})^{\frac{3}{2}}} \\
 K_4 &= \frac{dp_4}{d\tau} = \frac{i}{c} \frac{d(mc^2)}{d\tau} = \frac{i}{c} m_0 c^2 \frac{d}{d\tau} \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{i}{c} \frac{\frac{1}{2} m_0 \frac{dv^2}{d\tau}}{(1 - \frac{v^2}{c^2})^{\frac{3}{2}}} = \frac{i}{c} \vec{K} \cdot \vec{v}
 \end{aligned}$$



Back

Close



$$\begin{aligned}
 \nabla' \times \vec{\mathcal{E}}(\vec{r}', t') &= \nabla' \times \vec{\mathbf{E}}(\vec{r}' + \vec{v}t', t') & \vec{\mathcal{E}}(\vec{r}', t') &\equiv \vec{\mathbf{E}}(\vec{r}, t) \\
 &= \nabla \times \vec{\mathbf{E}}(\vec{r}, t) \\
 &= -\frac{\partial}{\partial t} \vec{\mathbf{B}}(\vec{r}, t) \\
 &= -\left(\frac{\partial}{\partial t'} - \vec{v} \cdot \nabla'\right) \vec{\mathbf{B}}(\vec{r}' + \vec{v}t', t') \Big|_{\text{第一组坐标变量固定}} \\
 &= -\frac{\partial}{\partial t'} \vec{\mathbf{B}}(\vec{r}' + \vec{v}t', t') \Big|_{\text{第一组坐标变量固定}} \\
 &= -\frac{d}{dt'} \vec{\mathbf{B}}(\vec{r}' + \vec{v}t', t') + \vec{v} \cdot \nabla' \vec{\mathbf{B}}(\vec{r}' + \vec{v}t', t') \\
 &= -\frac{\partial}{\partial t'} \vec{\mathcal{B}}(\vec{r}', t') + \vec{v} \cdot \nabla' \vec{\mathcal{B}}(\vec{r}', t') & \vec{\mathcal{B}}(\vec{r}', t') &\equiv \vec{\mathbf{B}}(\vec{r}, t)
 \end{aligned}$$

$$\vec{r} = \vec{r}' + \vec{v}t' \quad t = t' + \frac{\vec{v} \cdot \vec{r}'}{c^2} \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial t'} - \vec{v} \cdot \nabla' \quad \nabla' = \nabla + \frac{\vec{v}}{c^2} \frac{\partial}{\partial t}$$

黄色的部分是一阶狭义相对论修正



Back

Close



22/22



Back

Close